

Hyo-jeong (Tiffany) Lee

Gwangju, South Korea | hyojeonglee11190@gmail.com | Homepage | GitHub | LinkedIn

Summary

Computational neuroscientist working on **state-space analysis**, and **cognitive modeling**, **neural manifolds**. Experienced in multimodal neural datasets, cognitive experiments, and neural network modeling.

Education

Gwangju Institute of Science and Technology (GIST) <i>B.S. in Electrical Engineering & Computer Science, Minor in Biomedical Science & Engineering</i> ◦ GPA: 4.18/4.5, <i>Magna Cum Laude</i>	<i>Mar 2022 – Dec 2026</i>
University of California, Berkeley <i>Berkeley Global Access Program</i>	<i>Jan 2025 – May 2025</i>
Korea Advanced Institute of Science and Technology (KAIST) <i>Visiting Student</i>	<i>Aug 2024 – Dec 2024</i>
University of California, Berkeley <i>Summer Session Program</i>	<i>June 2023 – Aug 2023</i>

Research Experience

Neurophotronics Lab, GIST <i>Undergraduate Researcher (Advisor: Prof. Euiheon Chung)</i> ◦ Proposed a framework that rule inference emerges as a hierarchical computation grounded in dynamical neural motifs. ◦ Analyzed trial-to-trial shared variability in EEG and its alignment with the readout axis to account for age-related cognitive decline. ◦ Applied spectral graph theory on high-density multi-electrode array (HD-MEA) recordings and spiking neural networks (SNNs) to characterize mesoscale network dynamics.	<i>Gwangju, South Korea</i> <i>Sep 2025 – Present</i>
The Computation and Language Lab, UC Berkeley <i>Research Assistant (Advisor: Prof. Steven Piantadosi)</i> ◦ Showed that epsilon states form a low-dimensional, causally relevant subspace in recurrent neural network (RNN) and fMRI representations, supporting a task-general interpretation.	<i>Berkeley, United States</i> <i>May 2025 – Aug 2025</i>
Brain Machine Intelligence Lab, KAIST <i>Undergraduate Researcher (Advisor: Prof. Sang Wan Lee)</i> ◦ Supported study design, code review, and literature search for a project decoding prediction errors from EEG during model-based and model-free decision making.	<i>Daejeon, South Korea</i> <i>Sep 2024 – Dec 2024</i>
BioComputing Lab, GIST <i>Undergraduate Researcher (Advisor: Prof. Sung Chan Jun)</i> ◦ Designed and conducted a human cognitive task to investigate how label manipulation alters EEG neural correlates of deepfake discrimination.	<i>Gwangju, South Korea</i> <i>June 2024 – Aug 2024</i>
Neurophotronics Lab, GIST <i>Undergraduate Researcher (Advisor: Prof. Euiheon Chung)</i> ◦ Contributed to experimental setup and project conceptualization for HD-MEA neuronal network analysis.	<i>Gwangju, South Korea</i> <i>Dec 2022 – Feb 2024</i>

Publications

- Lee, H., & Piantadosi, S. T. (in preparation). Modeling Internal Representations of Task-Trained RNNs as Cognitive-State Automata.
- Lee, H., Ashiquzzaman, A., Kwon, H. S., Kim, Y. R., Kim, S. S., & Chung, E. (in preparation). Spectral Graph Theoretical Characterization of Cortical Neuronal Networks using HD-MEA.
- Ashiquzzaman, A., Lee, H., Lee, E., Kwon, H. S., Kim, Y. R., Kim, S. S., & Chung, E. (submitted). Cortical Neuronal Network Developmental-State Transitions Detection and Characterization with High-Density Multi-Electrode Array.

Conferences

- Lee, H., & Chung, E.** (2026, Nov 14-18). *Motif-based hierarchical rule inference in recurrent neural networks* [Poster]. Neuroscience (Society for Neuroscience), Washington D.C., United States.
- Lee, H., & Chung, E.** (2026, May 7-9). *Influence of Hierarchical Task Inference on Representational Dynamics in Healthy and Aging-like RNNs under Hidden-Rule Conditions* [Poster]. The Korean Society of Medical & Biological Engineering Spring Annual Meeting, Jeju, South Korea.
- Lee, H., & Chung, E.** (2025, Nov 8). *Age-Related Reorganization of Shared Neural Variability and Readout Alignment During Cognitive Control* [Poster]. The 19th International Symposium for Aging, Gwangju, South Korea.
- Lee, H., Ashiquzzaman, A., Kwon, H. S., Kim, Y. R., Kim, S. S., & Chung, E.** (2025, Nov 6-8). *Spectral Graph Theoretical Characterization of Cortical Neuronal Networks using HD-MEA* [Poster]. The Korean Society of Medical & Biological Engineering Fall Annual Meeting, Gimhae, South Korea.

Teaching Experience

- Computer Programming:** Teaching Assistant *Mar 2026 – Present*
- Open GIST Science Lab:** Programming Teaching Assistant *July 2026*
- Single Variable Calculus and Applications:** Teaching Assistant *Mar 2026 – June 2026*
- PIUM:** Science Tutoring Volunteer for the Suburbs *Sep 2022 – Nov 2022*

Extracurricular Activities

- Paper to Peer:** Mathematics Paper Reading Club *Aug 2026 – Present*
- Gravity:** Bio-imaging Mock Startup Project *June 2026 – Present*
- AIleoDreamy:** AI Newsletter Publishing Team *Nov 2024 – Aug 2025*
- Google Developer Student Clubs:** External Relations Leader *Sep 2023 – Feb 2024*
- Buddy Program:** Volunteer for International Freshmen *Sep 2023 – Dec 2023*
- Korean I-Corps:** AI Clothing Customization Mock Startup *Jun 2022 – Feb 2023*
- Asian Science Camp:** Representative of Korea *June 2022*

Skills

- Experiment:** Primary Neuron Culture, HD-MEA, Human Behavioral Tasks, EEG, Immunofluorescence, Calcium Imaging, fMRI analysis, Vibratome
- Tools:** ImageJ (Fiji), Git, Brainwave (3Brain), EEGLAB, L^AT_EX, UCINET
- Programming:** Python, MATLAB, C, R, Java
- Language:** Korean (Native), English (Fluent; TOEFL 113/120)

Awards & Honors

- National Scholarship for Science and Technology** *Mar 2022 – Dec 2026*
- Trainee Professional Development Award (Travel Grant)** *Aug 2026*
Society for Neuroscience
- GIST-Fellowship for Innovative Research in Science & Technology** *June 2026 - Aug 2026*
- Best Poster Award** *May 2026*
The Korean Society of Medical & Biological Engineering Spring Annual Meeting
- Best Poster Award** *Nov 2025*
The 19th International Symposium for Aging
- Study Abroad Program Scholarship** *Jan 2025 – Mar 2025*
- Academic Excellence Scholarship** *Sep 2024 – Dec 2024*
- AWS AI Award (1st place/200+ teams nationwide)** *Mar 2024*
AI4GOOD Hackathon
- Academic Excellence Scholarship** *Mar 2022 – Dec 2023*